

# From Independent Scientist to Novel Performer

A Small-Business Operating, Milestone, and Capitalization Plan for a Phase 1 LLM-Advised Robotic Whipple (ChemicalQDevice)

Draft 1.0

Repository v4.5.0

20 figures

12 milestones

SBIR Phase I

**\$306,000**

9 months, 5 milestones

SBIR Phase II

**\$1,300,000**

24 months, 7 milestones

Five-year programme

**\$3,500,000**

60 months, direct

CEO Kevin Kawchak , ChemicalQDevice | [kevink@chemicalqdevice.com](mailto:kevink@chemicalqdevice.com)  
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*Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.*

**Disclaimer:** This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v4.5.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in funding/capitalization-plan.

**Keywords** Small business innovation research; Novel performers; Capitalization plan; Milestone schedule; Indirect cost; 21 CFR part 54; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI

## Table of Contents

|                                                          |    |
|----------------------------------------------------------|----|
| Abstract                                                 | 2  |
| Executive Summary                                        | 2  |
| How to Read This Document                                | 2  |
| 1 The Novel-Performer Case                               | 6  |
| 1.1 Three Clauses, One Applicant                         | 6  |
| 1.2 Why the Pioneer Language Does Not Fit an Entity      | 7  |
| 1.3 Why the Novel-Performers Chapter Does Not Fit Either | 7  |
| 1.4 The Clause That Fits, and What It Is Worth           | 8  |
| 2 Placeholder                                            | 10 |
| 3 Placeholder                                            | 11 |
| 4 Placeholder                                            | 12 |
| 5 Placeholder                                            | 13 |
| 6 Placeholder                                            | 14 |
| 7 Placeholder                                            | 15 |
| 8 Placeholder                                            | 16 |
| 9 Placeholder                                            | 17 |
| 10 Placeholder                                           | 18 |
| 11 Placeholder                                           | 19 |

## Abstract

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In August 2026 the author filed ten funding applications for a Phase 1 first-in-human trial of an LLM-advised robotic pancreaticoduodenectomy with perioperative daraxonrasib, written throughout as an independent scientist [?]. Nine of the ten address mechanisms that fund a person or an institution. One, application 05, addresses the only mechanism in the set built for a company, and it was the shortest of the ten. This paper converts the author from independent scientist to small-business operator and rewrites that application as the document it should have been.

The clause the conversion turns on is neither the report's individual-scientist language nor its novel-performers chapter. It is the Small Business Innovation Research clause of *Science: A New Golden Age*, which appears in three separate chapters and is the only clause in the report written for a firm [?]. Three numbers govern what follows: \$306,000 for a nine-month Phase I, \$1,300,000 for a twenty-four-month Phase II, and the \$3,500,000 five-year direct programme the two are measured against [?]. Two more are derived: \$1,396,000 of direct work sits inside the \$1,606,000 award, and \$2,104,000 sits outside it.

What is offered is a costed twelve-milestone schedule. Every row carries an evidence artifact that exists as a file, a cost that sums into its phase, a stop condition stated before the work begins, and the funder mechanism that pays.

## Executive Summary

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**The clause.** *Science: A New Golden Age* names SBIR three times: in the Chapter I summary of recommendations, in Chapter III on deploying SBIR and STTR strategically to couple federally-seeded companies with the scientific enterprise, and in Chapter IV on SBIR opening doors for technician-founded ventures. Its novel-performers chapter describes work at tens of millions of dollars with teams of ten to a hundred people over half a decade. This company proposes \$1,606,000, 2.6 full-time equivalents, and 33 months.

**The entity.** ChemicalQDevice holds thirteen deposited assets outright and none under encumbrance: two protocols, an IND package, a simulation stack, a verification suite, and a public repository. It licenses three things, none exclusively. It has contracted for nothing. Seven instruments are absent, and not one can be obtained by the company alone.

**The two prices.** The identical four-layer budget costs \$3,500,000 of direct work over sixty months and \$1,396,000 of direct work over thirty-three. The award that carries the second is \$1,606,000, a 15.0 percent overhead and fee load against an SBIR allowance of 40 percent that the company does not claim. Routed through a university at a 57 percent facilities and administrative rate, the same direct work would cost a funder \$2,137,000.

**The bridge.** Federal money of \$1,606,000 sits below a firewall; contributed drug, theatre and pathology sit unpriced in the middle; and \$5,900,000 of private capital sits above, raised only after a milestone closes. That is 3.67 to one on cash alone, against the annexed FY 2028 target of at least three to one [?].

**The milestones.** Twelve rows, five in Phase I summing to \$306,000 and seven in Phase II summing to \$1,300,000. Each is closed by three concurrent evidence branches that must join before an audit can compare them.

**What the conversion costs.** The individual-scientist framing that made all ten prior applications distinctive is abandoned. Capitalization detail the author might prefer private is published permanently under a DOI. Three of the ten prior recipients will read a company plan as off-mechanism. Those four concessions are stated in §9 and are not softened anywhere.

## How to Read This Document

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The document is written so that four different readers can each open it at a different page and find the thing they came for within two pages.

| Reader                   | Open at       | Then                 | The question this reader is actually asking                       |
|--------------------------|---------------|----------------------|-------------------------------------------------------------------|
| SBIR program director    | §3, Table 8   | §5, Tables 13 and 14 | Does the award buy a defensible gate, and does the budget sum?    |
| ARPA-H programme manager | §5, Figure 13 | §1, Figure 1         | Is there a milestone I can hold someone to, and a stop condition? |
| Private investor         | §4, Table 12  | §2, Table 6          | What do I own, when may I own it, and what is already built?      |
| UC San Diego intake      | §8, Figure 19 | §7, Figure 18        | What is being asked of this institution, and by when?             |

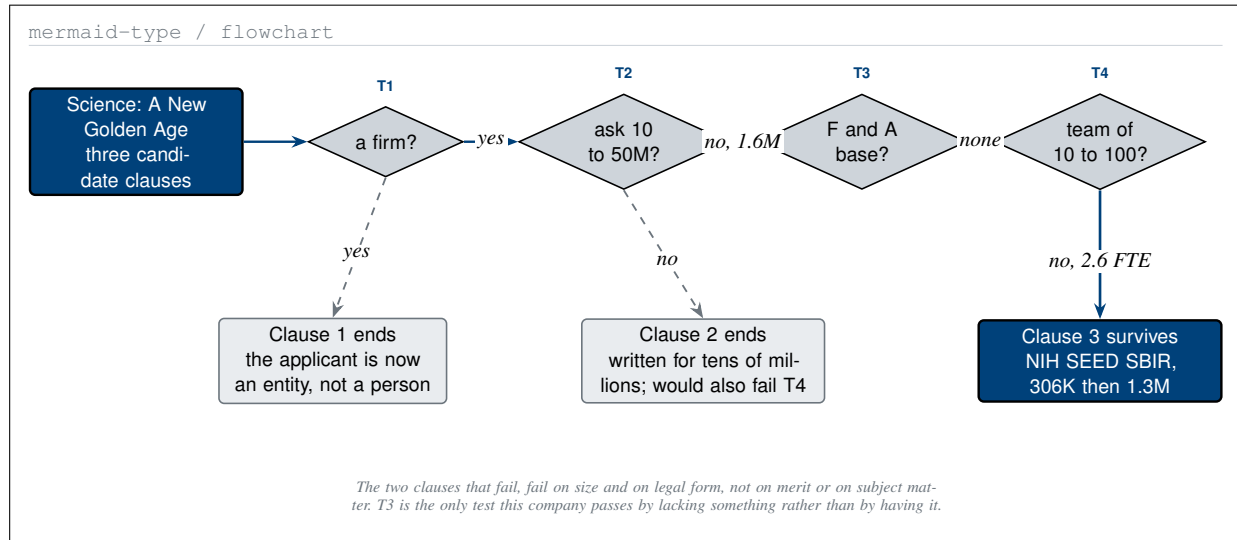
*Four readers and the section each should open first. No reader is asked to read the document front to back, and every entry point is a table or a figure rather than a paragraph.*

| Fig | §  | Platform | Source directory | The question it answers                                               |
|-----|----|----------|------------------|-----------------------------------------------------------------------|
| 1   | 1  | Mermaid  | mermaid/         | Which clause does a company with no indirect-cost base qualify under? |
| 2   | 1  | D2       | d2/              | What kind of institution is a 2.6 FTE firm, on the report's own axes? |
| 3   | 1  | Graphviz | graphviz/        | What does the same direct work cost under three overhead regimes?     |
| 4   | 2  | Diagrams | diagrams-pyhton/ | What is the shape of what the company holds today?                    |
| 5   | 2  | D2       | d2/              | What is owned, licensed, contracted, and absent, with terms?          |
| 6   | 2  | PlantUML | plantuml/        | What may the sponsor do alone, and what needs an institution?         |
| 7   | 3  | Mermaid  | mermaid/         | What must hold at month nine for Phase II to begin?                   |
| 8   | 3  | D2       | d2/              | What does the same programme cost under two mechanisms?               |
| 9   | 3  | Graphviz | graphviz/        | Which work packages does the money reach, and which not?              |
| 10  | 4  | PlantUML | plantuml/        | Which capital positions may be held while a participant is on study?  |
| 11  | 4  | D2       | d2/              | Where does every dollar come from, and what separates the tiers?      |
| 12  | 4  | Mermaid  | mermaid/         | In what order is a private round executed during enrollment?          |
| 13  | 5  | Mermaid  | mermaid/         | When does each milestone open, close, and produce its artifact?       |
| 14  | 5  | Graphviz | graphviz/        | What has to fail, and in what combination, to stop the programme?     |
| 15  | 5  | PlantUML | plantuml/        | What do three parties do at once to close one milestone?              |
| 16  | 6  | D2       | d2/              | Which published quantities can a funder check before committing?      |
| 17  | 6  | Graphviz | graphviz/        | Where does each published number land in a filing?                    |
| 18  | 7  | Diagrams | diagrams-pyhton/ | Who is employed, who is contracted, who is contributed?               |
| 19  | 8  | Mermaid  | mermaid/         | What did the July and August 2026 contacts actually unlock?           |
| 20  | 10 | Diagrams | diagrams-pyhton/ | What survives if the programme stops, and who holds it?               |

*The twenty figures, their section, their platform, the directory that carries each specification, and the one question each answers. Five mermaid, three plantuml, five d2, three diagrams, four graphviz, by purpose not quota.*

| Tab | §  | Contents                                         | The question it answers                                        |
|-----|----|--------------------------------------------------|----------------------------------------------------------------|
| 1   | 0  | Four readers, four entry points                  | Which section should this reader open first?                   |
| 2   | 0  | Figure index, twenty rows                        | Where is each figure and what does it answer?                  |
| 3   | 0  | Table index, nineteen rows                       | Where is each table and what does it answer?                   |
| 4   | 1  | Three clauses against four eligibility tests     | Which clause survives the filter, and why?                     |
| 5   | 1  | Three overhead regimes on one direct base        | What is the premium for routing through an institution?        |
| 6   | 2  | The owned register, thirteen rows with DOIs      | What exists today and where is it deposited?                   |
| 7   | 2  | The absent register, seven rows with signatories | What is missing and who alone can supply it?                   |
| 8   | 3  | The four-layer budget under two prices           | What does the SBIR route reach, and what does it not?          |
| 9   | 3  | Phase I line items, eight rows to \$306,000      | What does nine months of Phase I actually buy?                 |
| 10  | 3  | Phase II line items, ten rows to \$1,300,000     | What does twenty-four months of Phase II buy?                  |
| 11  | 4  | The §54.2 and §54.4 disclosure triggers          | What must be disclosed, at what threshold, on which form?      |
| 12  | 4  | The capitalization table across five stages      | Who owns what, and does the SBIR test still pass?              |
| 13  | 5  | The twelve milestones, costed and dated          | What is the milestone, when, and what does it cost?            |
| 14  | 5  | Artifact, stop condition, funder mechanism       | What proves it, what halts it, and who pays?                   |
| 15  | 6  | Six checkable quantities with limitations        | What can a reviewer verify against a published source?         |
| 16  | 6  | Cost and schedule against industry benchmarks    | What did the prior work cost, and what does it cost elsewhere? |
| 17  | 7  | The 2.6 FTE by employment status                 | Who is on the payroll and who is not?                          |
| 18  | 9  | Risks, likelihood, and the mitigation held       | What could go wrong and what is already in place?              |
| 19  | 11 | Abbreviations, two column pairs                  | What does this term mean in this paper?                        |

*The nineteen tables, their section, their contents, and the one question each answers. Every table is set to the full body measure with ragged-right fixed columns, so no cell shows a stretched interword gap.*



*Three clauses of one report, four eligibility tests, and the only clause a San Diego firm with no indirect-cost base survives. The two that fail, fail on size, not on merit or on subject matter.*

## 1 The Novel-Performer Case

### 1.1 Three Clauses, One Applicant

*Science: A New Golden Age* contains three clauses under which a Phase 1 pancreatic cancer trial could plausibly be funded, and they are not interchangeable [?].

The first is the individual-scientist language of Chapter I, which asks the roughly \$200 billion annual federal research and development portfolio to stop deferring to “the same incumbents that consume the funding”. The ten prior applications were written under it [?].

The second is the chapter section headed NOVEL PERFORMERS, which describes “a vast middle ground of mid-scale science”, problems “requiring tens of millions of dollars, coordinated teams of ten to a hundred people, and timelines of half a decade”.

The third is the Small Business Innovation Research clause, and it is the only one of the three that appears in more than one chapter. The Chapter I summary of recommendations asks agencies to “focus Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs to build strategic capabilities”. Chapter III states that “SBIR and STTR programs can be deployed strategically to advance new scientific and technological capabilities, coupling federally-seeded companies with the scientific enterprise”. Chapter IV states that “Programs like SBIR open doors for technician-founded ventures, directing resources toward ideas that do not always start with a dissertation”.

| Test | Question                                      | Basis in the report                                              | ChemicalQDevice, August 2026          |
|------|-----------------------------------------------|------------------------------------------------------------------|---------------------------------------|
| T1   | Is the applicant a firm rather than a person? | Chapter IV, SBIR opens doors for technician-founded ventures     | Yes, a California corporation         |
| T2   | Is the ask ten to fifty million?              | Chapter II, novel performers is written for tens of millions     | No, \$1,606,000 over 33 months        |
| T3   | Is there an indirect-cost base to recover?    | Chapter II, deference to the incumbents that consume the funding | No negotiated rate agreement exists   |
| T4   | Is the team ten to a hundred people?          | Chapter II, coordinated teams of ten to a hundred                | No, 2.6 FTE at full Phase II staffing |

*The four tests, the sentence in the report that each is drawn from, and this company's answer. Clause 1 fails T1, clause 2 fails T2 and T4, and clause 3 is the only one that survives all four.*

## 1.2 Why the Pioneer Language Does Not Fit an Entity

What changed between the ten applications and this one is the applicant. A person-based mechanism funds a named scientist and follows that person's judgement for five to seven years. ChemicalQDevice is a California corporation with one officer, and a corporation is not a subject that mechanism has a way to fund. Nothing about the science changed; the legal form of the applicant did.

That is a real loss and it is worth naming here rather than in a footnote. The individual-scientist framing is what made all ten prior applications distinctive, and it is the framing the report itself rewards most explicitly. Abandoning it is the first of four costs the conversion carries, and all four are set out in §9 without mitigation.

## 1.3 Why the Novel-Performers Chapter Does Not Fit Either

The novel-performers chapter is precise about scale, and the precision is what excludes this company. It describes problems requiring tens of millions of dollars, teams of ten to a hundred people, and timelines of half a decade, and it names focused research organizations and the NSF TIP X-Labs as the vehicles built for them. Set against that: \$1,606,000, 2.6 full-time equivalents, and 33 months.

The chapter describes a real gap in the federal portfolio. This company does not fill it, and claiming otherwise would be exactly the kind of overreach the report is written against.

Two well suited, three partly suited, two less suited. The two well-suited rows are public-goods data and infrastructure, and mission-driven public-good science, and both are well suited for the same structural reason: every one of the twelve milestone artifacts in §5 is deposited publicly by plan, and the programme has one mission with a stop condition and a closure budget.

A firm claiming to be well suited to all seven would be claiming to be a university, a corporate laboratory, a federal laboratory and a focused research organization at once. The report's own table exists to make that claim impossible to state.

d2-type / true grid

| Activity                              | University    | Corporate lab | Federal lab   | New institutions | SBIR firm 2.6 FTE |
|---------------------------------------|---------------|---------------|---------------|------------------|-------------------|
| Curiosity driven, investigator led    | Well suited   | Less suited   | Partly suited | By design        | Partly suited     |
| Larger scale, engineering intensive   | Less suited   | Partly suited | Partly suited | By design        | Less suited       |
| Long horizon platform and tools       | Less suited   | Partly suited | Well suited   | By design        | Partly suited     |
| Public goods data and infrastructure  | Partly suited | Less suited   | Partly suited | By design        | Well suited       |
| Mission driven, public good science   | Partly suited | Less suited   | Partly suited | By design        | Well suited       |
| Proprietary product development       | Less suited   | Well suited   | Less suited   | By design        | Partly suited     |
| Workforce training and apprenticeship | Well suited   | Partly suited | Less suited   | By design        | Less suited       |

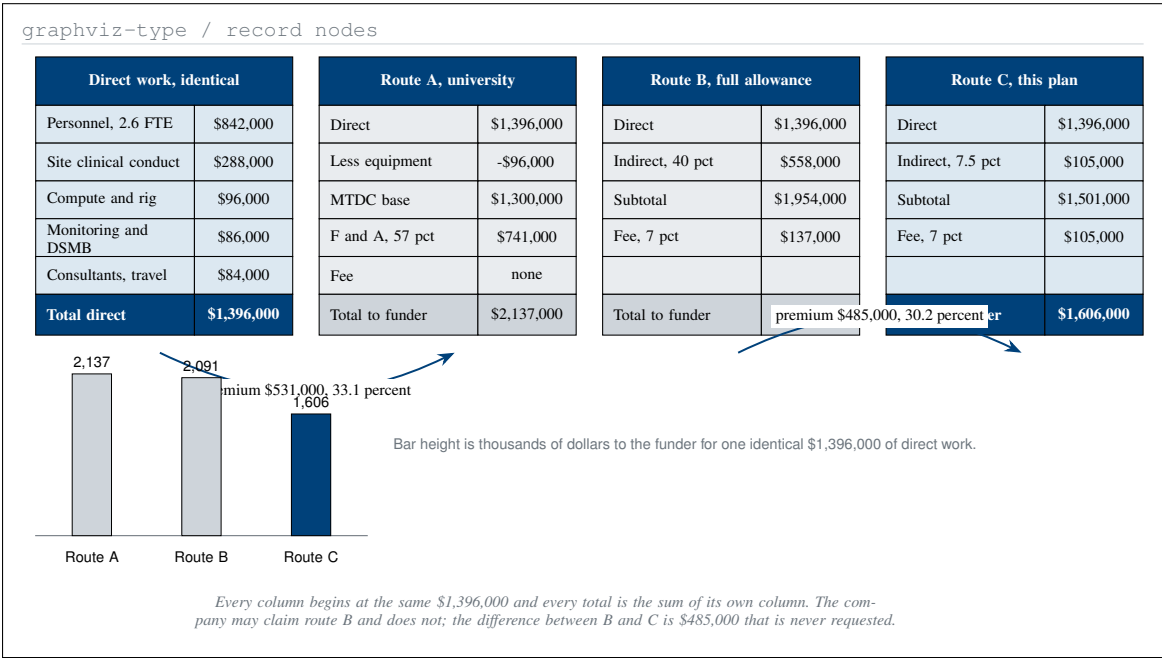
☐ Well or partly suited, the report's own scoring

☐ Less suited

☒ Well suited, this company

The rule at the right marks where the report's own columns end and the column this paper adds begins. The four left-hand columns are the report's scoring restated in words, because its plus, tilde and cross glyphs are not legible at this measure.

The report's own seven activities, its four institutional forms, and a fifth column it does not carry. An SBIR firm of 2.6 people is well suited to two of the seven and unsuited to two others.



The same 1,396,000 of direct work priced three ways. A university F and A rate at 57 percent adds 741,000, the full SBIR allowance adds 695,000, and the rate this plan claims adds 210,000 in all.

1.4 The Clause That Fits, and What It Is Worth

ChemicalQDevice has no negotiated indirect-cost rate agreement, no facilities to recover, and no administrative layer to fund. That is a structural fact about a 2.6 person firm rather than a concession, and it has a price a program officer with a fixed appropriation can read directly.



| Line                        | Route A,<br>university | Route B, full<br>allowance | Route C, this<br>plan |
|-----------------------------|------------------------|----------------------------|-----------------------|
| Direct costs                | \$1,396,000            | \$1,396,000                | \$1,396,000           |
| Equipment removed from base | -\$96,000              | not applicable             | not applicable        |
| Overhead base               | \$1,300,000<br>MTDC    | \$1,396,000 direct         | \$1,396,000<br>direct |
| Overhead rate               | 57 percent             | 40 percent                 | 7.5 percent           |
| Overhead                    | \$741,000              | \$558,000                  | \$105,000             |
| Fee at 7 percent            | none                   | \$137,000                  | \$105,000             |
| Total to the funder         | \$2,137,000            | \$2,091,000                | \$1,606,000           |
| Premium over route C        | \$531,000, 33.1<br>pct | \$485,000, 30.2 pct        | zero                  |

*The same direct work under three overhead regimes. The 57 percent rate is a typical negotiated university rate; the 40 percent is the SBIR allowance available without a rate agreement; the 7.5 percent is what this firm documents.*

The 7.5 percent rate covers insurance, audit and accounting, and nothing else, because there is nothing else to cover [?]. The company could claim the 40 percent allowance without documentation and does not [?]. For a program officer holding a fixed appropriation the consequence is arithmetic rather than rhetorical: the same trial, run by the same person, costs \$531,000 less than it would through an institution and \$485,000 less than this mechanism permits.

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## 2 Placeholder

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### 3 Placeholder

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## 4 Placeholder

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## 6 Placeholder

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## 9 Placeholder

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## 11 Placeholder

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